

Solving $Ax = b$: Row Reduced Form R

Now the client wants a specific target. Four short steps on one simplified matrix $[R \mid d]$ tell you whether it can be hit, give one portfolio that hits it, and then every portfolio that does.

PART 1 Index

Note 2 of the pair: $Ax = b$, solving for a specific target. (Note 1 is Lesson 7, $Ax = 0$.)

1. Conditions and important concepts

12 short definitions: $[A \mid b]$, $[R \mid d]$, zero rows, contradiction, free variables, the three possibilities, particular and complete solutions, rank conditions.

2. Procedure for $Ax = b$

The 7 steps, in order.

3. 2×2 examples

Exactly one, infinitely many, none.

4. 3×3 examples

Exactly one, infinitely many, none.

5. Final decision tree

Contradiction? Free variables? The answer.

6. Connection between $Ax = 0$ and $Ax = b$

Every solution = one solution + a null space solution. With a target tester.

7. Final memory table

$Ax = 0$ and $Ax = b$ side by side.

METHODS IN THIS LESSON

Augmented matrix $[A \mid b]$

$[A \mid b] \rightarrow [R \mid d]$

Contradiction test

Particular solution

Complete solution $x_p + x_n$

Rank conditions

PART 2 Prerequisites

Click any link to review it, then come back.

- Note 1: $Ax = 0$ (Lesson 7): R, pivots, free variables, special solutions, N.
- $Ax = b$ is solvable exactly when b is in $C(A)$ (Lesson 6).

THE FUNDS USED IN THE EXAMPLES

Nifty (1, 12, 1), Nifty ETF (1, 12, 1), bonds (1, 7, 0), balanced (1, 9.5, 0.5), midcap (1, 15, 1.2): what ₹1 lakh of each does to (budget, return, exposure). A target asks for a budget (₹ lakh), a return (₹ thousand) and an exposure.

1. Conditions and important concepts

1. What is $Ax = b$?

$Ax = b$ means:

Find the portfolio x that produces the specific target b .

2. What is b ?

b is the target. For example:

$$b = \begin{bmatrix} B \\ R \\ E \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \\ 7 \end{bmatrix} \quad \begin{array}{l} B = \text{budget ₹10 lakh} \\ R = \text{return ₹105 thousand} \\ E = \text{exposure 7} \end{array}$$

3. Augmented matrix $[A \mid b]$

METHOD · AUGMENTED MATRIX $[A \mid B]$

Attach the target b to A as an extra column. The vertical line separates the matrix (left) from the target (right).

For Nifty, bonds, balanced and the target above:

$$[A \mid b] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 12 & 7 & 9.5 & 105 \\ 1 & 0 & 0.5 & 7 \end{array} \right]$$

4. What happens to $[A \mid b]$?

METHOD · ROW REDUCE $[A \mid B] \rightarrow [R \mid D]$

Row reduce the **entire** augmented matrix. The same row operations are applied to A and to b . The left part becomes R , exactly as in Note 1.

The row operations are Note 1's (3×3 example). Here is what each one does to the target column:

Row operation	Target column
Row 2 $- 12 \times$ row 1	$105 - 12(10) = -15$
Row 3 $- 1 \times$ row 1	$7 - 10 = -3$
Row 3 $- 0.2 \times$ row 2	$-3 - 0.2(-15) = -3 + 3 = 0$
Row 2 $\div (-5)$	$-15 / (-5) = 3$
Row 1 $- 1 \times$ row 2	$10 - 3 = 7$

5. What is d ?

After row reduction, the original target b becomes a simplified target column called d :

$$[R \mid d] = \left[\begin{array}{ccc|c} 1 & 0 & 0.5 & 7 \\ 0 & 1 & 0.5 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad d = \begin{bmatrix} 7 \\ 3 \\ 0 \end{bmatrix}$$

6. Zero row

A zero row looks like $[0 \ 0 \ 0 \mid \text{something}]$. There are **two** possibilities.

Condition 1: $[0 \ 0 \ 0 \mid 0]$. This means

$$\boxed{0 = 0}$$

This is fine. There is **no contradiction**.

Condition 2: $[0 \ 0 \ 0 \mid \text{non-zero}]$. For example (target $(10, 105, 5)$, see the 3×3 examples):

$$\boxed{0 = -2}$$

This is impossible: whatever x_1, x_2, x_3 are, the left side is always zero. Therefore **no solution**.

7. Contradiction condition

RULE · CONTRADICTION TEST

If after row reduction we get a row saying $0 = \text{non-zero}$, then there is **no solution**.

8. Free variable condition

If there are **no contradictions**:

No free variables $\rightarrow$ exactly ONE solution

At least one free variable $\rightarrow$ INFIN

9. The three possibilities

Contradiction $\rightarrow$ NO SOLUTION

No contradiction + no free variables $\rightarrow$ EXACTLY ONE SOLUTION

No contradiction + free variables $\rightarrow$ INFINITELY MANY SOLUTIONS

10. Particular solution

METHOD · PARTICULAR SOLUTION x_p

A particular solution is simply **one** solution that satisfies $Ax = b$. To find an easy one: set all free variables $= 0$, then solve for the pivot variables (each pivot variable $=$ its row's d).

For the $[R \mid d]$ above: free $x_3 = 0$, so $x_1 = 7, x_2 = 3$: $x_p = (7, 3, 0)$, ₹7 lakh Nifty and ₹3 lakh bonds.

11. Complete solution

Once we have one particular solution x_p , we can add any null space solution x_n :

$$x = x_p + x_n$$

Why? Because $Ax_p = b$ and $Ax_n = 0$. Therefore

$$A(x_p + x_n) = Ax_p + Ax_n = b + 0 = b.$$

RULE · COMPLETE SOLUTION

Every solution $=$ one solution $+$ a null space solution. (And nothing else: if x is any solution, $x - x_p$ gives $b - b = 0$, so it is a null space solution.)

For the example: $x_n = t(-0.5, -0.5, 1)$ from Note 1, so $x = (7, 3, 0) + t(-0.5, -0.5, 1)$.

12. Rank conditions

Suppose A has m rows, n columns and rank r . Then:

number of zero rows = $m - r$

number of free variables = $n - r$

RULE · RANK CONDITIONS

If $r = m$: no zero rows, so never a contradiction: every target is reachable.

If $r = n$: no free variables: at most one solution.

If $r = m = n$: no zero rows **and** no free variables: exactly one solution for every target.

Funds (3 rules)	m	n	r	What it means
Nifty, bonds, midcap	3	3	3	$r = m = n$: exactly one solution for every target
Nifty, bonds	3	2	2	$r = n$: at most one solution
Nifty, bonds, midcap, balanced	3	4	3	$r = m$: every target reachable (infinitely many ways)
Nifty, bonds, balanced	3	3	2	neither: none or infinitely many

2. Procedure for $Ax = b$

Step	What to do
1	Build $[A \mid b]$.
2	Row reduce: $[A \mid b] \rightarrow [R \mid d]$.
3	Check the zero rows. $0 = \text{non-zero} \rightarrow \text{no solution}$ (stop). $0 = 0 \rightarrow \text{continue}$.
4	Find the pivot and free variables.
5	No free variables $\rightarrow$ exactly one solution. Free variables $\rightarrow$ infinitely many .
6	To find one solution: set all free variables $= 0$ and solve for the pivot variables.
7	To find every solution: keep the free variables arbitrary (equivalently, add the null space solutions).

3. 2×2 examples

Budget and return only.

Example A: exactly one solution

Nifty and bonds, target $(10, 105)$:

$$\left[\begin{array}{cc|c} 1 & 1 & 10 \\ 12 & 7 & 105 \end{array} \right]$$

- Row 2 $- 12 \times$ row 1: $(0, 7 - 12 = -5 \mid 105 - 120 = -15)$.
- Row 2 $\div (-5)$: $(0, 1 \mid 3)$.
- Row 1 $-$ row 2: $(1, 0 \mid 10 - 3 = 7)$.

$$[R \mid d] = \left[\begin{array}{cc|c} 1 & 0 & 7 \\ 0 & 1 & 3 \end{array} \right]$$

No contradictions. No free variables. Therefore **exactly one solution**:

$$\boxed{x_1 = 7, \quad x_2 = 3} \quad x = \begin{bmatrix} 7 \\ 3 \end{bmatrix}$$

Check: $7 + 3 = 10 \checkmark, 84 + 21 = 105 \checkmark$.

Example B: infinitely many solutions

Nifty and its twin ETF, target (10, 120):

$$\left[\begin{array}{cc|c} 1 & 1 & 10 \\ 12 & 12 & 120 \end{array} \right] \xrightarrow{\text{row 2} - 12 \times \text{row 1}} \left[\begin{array}{cc|c} 1 & 1 & 10 \\ 0 & 0 & 0 \end{array} \right]$$

(Row 2: $12 - 12 = 0$, $12 - 12 = 0$, $120 - 120 = 0$.) There is no contradiction ($0 = 0$). But x_2 is free. Therefore **infinitely many solutions**.

Let $x_2 = c$. Then $x_1 + c = 10$, so $x_1 = 10 - c$:

$$x = \begin{bmatrix} 10 - c \\ c \end{bmatrix} \quad \text{any split of ₹10 lakh between the twins}$$

Example C: no solution

The twins again, but target (10, 105):

$$\left[\begin{array}{cc|c} 1 & 1 & 10 \\ 12 & 12 & 105 \end{array} \right] \xrightarrow{\text{row 2} - 12 \times \text{row 1}} \left[\begin{array}{cc|c} 1 & 1 & 10 \\ 0 & 0 & -15 \end{array} \right]$$

The second row says

$$0 = -15$$

Impossible. Therefore **no solution**. (Two copies of a 12% fund can't earn 10.5%.)

4. 3 × 3 examples

Example A: exactly one solution

Nifty, bonds, midcap, target (10, 111, 7.4):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 12 & 7 & 15 & 111 \\ 1 & 0 & 1.2 & 7.4 \end{array} \right]$$

- Row 2 $- 12 \times$ row 1: $(0, -5, 3 \mid 111 - 120 = -9)$.
- Row 3 $- 1 \times$ row 1: $(0, -1, 0.2 \mid 7.4 - 10 = -2.6)$.
- Row 3 $- 0.2 \times$ row 2: $(0, 0, 0.2 - 0.6 = -0.4 \mid -2.6 + 1.8 = -0.8)$.
- Row 3 $\div (-0.4)$: $(0, 0, 1 \mid 2)$.
- Row 2 $- 3 \times$ row 3: $(0, -5, 0 \mid -9 - 6 = -15)$; then $\div (-5)$: $(0, 1, 0 \mid 3)$.

- Row 1 – row 2 – row 3: $(1, 0, 0 \mid 10 - 3 - 2 = 5)$.

$$[R \mid d] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

3 pivots, 0 free variables, no contradiction. Therefore **exactly one solution**:

$$x = \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix} \quad \text{₹5L Nifty, ₹3L bonds, ₹2L midcap}$$

Check: $5 + 3 + 2 = 10 \checkmark$, $60 + 21 + 30 = 111 \checkmark$, $5 + 0 + 2.4 = 7.4 \checkmark$.

Example B: infinitely many solutions

Nifty, bonds, balanced, target $(10, 105, 7)$. Concepts 4 and 5 above already reduced it:

$$[R \mid d] = \left[\begin{array}{ccc|c} 1 & 0 & 0.5 & 7 \\ 0 & 1 & 0.5 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

No contradiction ($0 = 0$). But x_3 is free. Therefore **infinitely many solutions**.

Let $x_3 = t$. Then $x_1 + 0.5t = 7$, so $x_1 = 7 - 0.5t$; and $x_2 + 0.5t = 3$, so $x_2 = 3 - 0.5t$:

$$x = \begin{bmatrix} 7 - 0.5t \\ 3 - 0.5t \\ t \end{bmatrix}$$

Check $t = 2$: $(6, 2, 2)$. Budget $6 + 2 + 2 = 10 \checkmark$, return $72 + 14 + 19 = 105 \checkmark$, exposure $6 + 0 + 1 = 7 \checkmark$.

Example C: no solution

The same funds, but target $(10, 105, 5)$. The same row operations act on the new target column: $105 - 120 = -15$; $5 - 10 = -5$; $-5 - 0.2(-15) = -5 + 3 = -2$; $-15/(-5) = 3$; $10 - 3 = 7$.

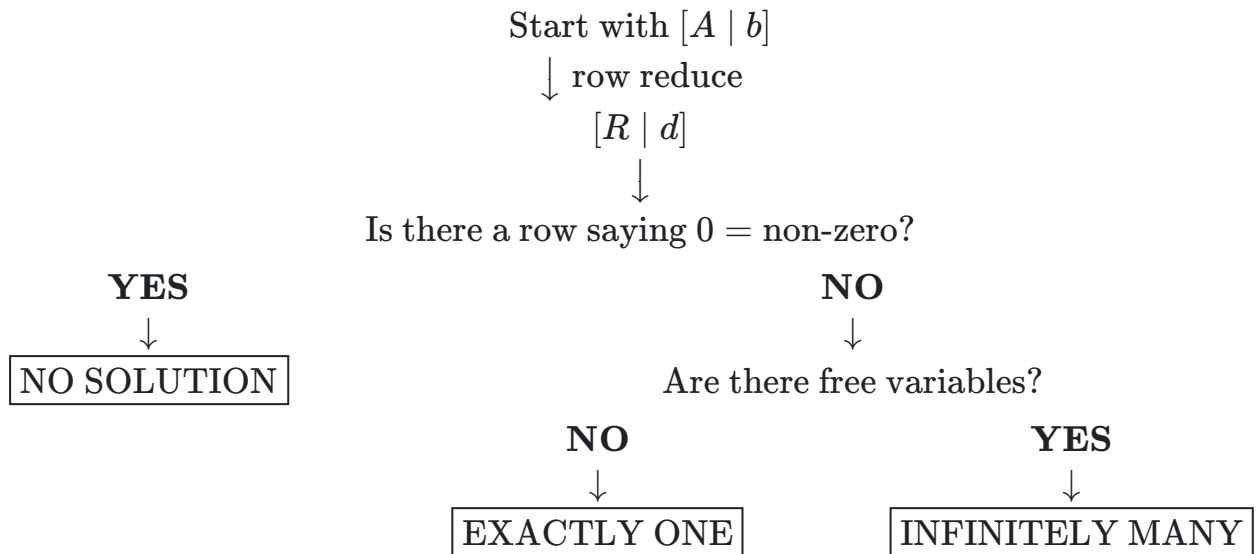
$$[R \mid d] = \left[\begin{array}{ccc|c} 1 & 0 & 0.5 & 7 \\ 0 & 1 & 0.5 & 3 \\ 0 & 0 & 0 & -2 \end{array} \right]$$

The last row says

$$0 = -2$$

Impossible. Therefore **no solution**. (For these funds, ₹10 lakh earning ₹105 thousand forces exposure 7; asking for 5 contradicts it.)

5. Final decision tree



Contradiction?	Free variables?	Answer	Examples above
yes	(doesn't matter)	no solution	2×2 C, 3×3 C
no	no	exactly one	2×2 A, 3×3 A
no	yes	infinitely many	2×2 B, 3×3 B

Always check for a contradiction **first**: 2×2 B and C use the same funds (with a free variable), but in C the contradiction wins.

6. Connection between $Ax = 0$ and $Ax = b$

	$Ax = 0$ (Note 1)	$Ax = b$ (Note 2)
Find	all inputs that produce ZERO	all inputs that produce a SPECIFIC TARGET
Main concepts	null space	particular solution + null space

The complete solution is

$$\boxed{x = x_p + x_n} \quad \begin{array}{l} x_p = \text{one solution that reaches } b \\ x_n = \text{any null space solution} \end{array}$$

because $Ax_p = b$ and $Ax_n = 0$, so $A(x_p + x_n) = b$.

On the four-fund menu (Note 1's bigger example)

Nifty, ETF, bonds, balanced, target $(10, 105, 7)$. The same five row operations as Note 1, applied to the target column too, give:

$$[R \mid d] = \left[\begin{array}{cccc|c} 1 & 1 & 0 & 0.5 & 7 \\ 0 & 0 & 1 & 0.5 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

No contradiction; free variables x_2 (ETF), x_4 (balanced). Free = 0 gives $x_p = (7, 0, 3, 0)$. Note 1 gave the null space: $x_n = c s_1 + d s_2$. So:

$$x = \underbrace{\begin{bmatrix} 7 \\ 0 \\ 3 \\ 0 \end{bmatrix}}_{x_p} + c \underbrace{\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}}_{x_n} + d \underbrace{\begin{bmatrix} -0.5 \\ 0 \\ -0.5 \\ 1 \end{bmatrix}}_{x_n}$$

Check $c = 3, d = 2$: $(7 - 3 - 1, 3, 3 - 1, 2) = (3, 3, 2, 2)$. Budget $3 + 3 + 2 + 2 = 10 \checkmark$, return $36 + 36 + 14 + 19 = 105 \checkmark$, exposure $3 + 3 + 0 + 1 = 7 \checkmark$.

In money terms: once one portfolio hits the target, swapping Nifty for the ETF or building balanced from Nifty and bonds keeps it hit.

Try it: any target, any free amounts

Budget B Return R Exposure E

ETF c Balanced d

Zero-row test: $7B + 5E - R = 7(10) + 5(7) - 105 = 0$.

$[R \mid d]$ has last column $d = (7, 3, 0)$. **One answer** (free amounts = 0): $x_p = (7, 0, 3, 0)$.

Every answer (ETF = c , balanced = d): $x = x_p + 2 s_1 + 2 s_2 = (4, 2, 2, 2)$ (₹ lakh in Nifty, ETF, bonds, balanced).

Budget: $1(4) + 1(2) + 1(2) + 1(2) = 10$ ✓
 Return: $12(4) + 12(2) + 7(2) + 9.5(2) = 105$ ✓
 Exposure: $1(4) + 1(2) + 0(2) + 0.5(2) = 7$ ✓

Hits the target. Change c and d: it still does.

7. Final memory table

	$Ax = 0$	$Ax = b$
Goal	Find inputs that give ZERO.	Find inputs that give TARGET b .
Row reduce	$A \rightarrow R$	$[A \mid b] \rightarrow [R \mid d]$
Free variables	Create special solutions.	Decide between one and infinitely many.
Key objects	Special solutions form the null space; nullspace matrix N .	Particular solution x_p (free variables = 0).
Key relationship	$AN = 0$	$x = x_p + x_n$
Contradiction	never ($0 = 0$ always)	$0 = \text{non-zero} \rightarrow \text{no solution}$
Outcomes	No free variables $\rightarrow$ only $x = 0$. Free variables $\rightarrow$ infinitely many.	No contradiction + no free variables $\rightarrow$ exactly one. No contradiction + free variables $\rightarrow$ infinitely many.